Timing Matters: A Comprehensive Meta-Analysis on the Optimal Period for Cranioplasty After Severe Traumatic Brain Injury

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BACKGROUND AND OBJECTIVES: Craniectomy is a commonly performed procedure in severe traumatic brain injury (TBI) to control intracranial hypertension and reduce mortality. The optimal timing for cranioplasty after craniectomy remains a topic of debate. The aim of this study was to investigate the ideal timing for cranioplasty after severe TBI, focusing on complications associated with temporal aspects and materials used in cranioplasty.

METHODS: Multiple databases, including PubMed, Embase, Cochrane, and Web of Science, were searched for studies reporting on the subject. Inclusion criteria involved randomized and observational studies comparing early and late cranioplasty procedures. Quality assessment was performed using the Methodological Index for Non-Randomized Studies scale. Results were pooled in a single-arm meta-analysis and presented as mean and 95% confidence interval.

RESULTS: Early cranioplasty was associated with a lower likelihood of subdural effusion (odds ratio [OR] 0.3735 [0.1643; 0.8490], $P = .0187$). However, no differences were detected for the presence of infection between both groups (OR 0.7460 [0.2065; 2.6945], $P = .6548$). Cranioplasties performed within or equal to 3 months from the TBI incident were associated with a reduced likelihood of minor complications (OR 0.4471 [0.2467; 0.8102], $P = .0080$). Paradoxically, this time frame exhibited an elevated risk of hydrocephalus (OR 3.2035 [1.4860; 6.9059], $P = .0030$), as well as total complications (OR 1.4190 [1.0453; 1.9262] $P = .0248$).

CONCLUSION: This comprehensive review highlights the current lack of consensus on the optimal timing for cranioplasty after severe TBI. Early cranioplasty, within 35 days, showed a reduced risk of subdural effusion, but no disparity in the odds of infection was found. A reduced incidence of minor complications was observed in the earlier approach, however with an elevated risk for hydrocephalus and total complications. The association between timing and complications, including hydrocephalus, highlights the need for further research and standardization in this critical aspect of TBI management.

KEY WORDS: Cranioplasty, Traumatic brain injury, Complications, Humans, Meta-analysis, Treatment outcome

Craniectomy (DC) is one of the most commonly performed procedures in the setting of severe traumatic brain injury (TBI) to control intracranial hypertension and reduce mortality in these patients. This situation generates a significant number of patients requiring cranioplasty. Cranioplasty (CP) is a prominent method for addressing skull deformities for cosmetic, protection, and, in some cases, functional improvement purposes.[1-3] The therapeutic efficacy is underscored by evident improvements in neuropsychological and cognitive functions after CP and notable enhancements in cerebral blood flow and cerebrospinal fluid circulation.[4-7]

Although statistically significant associations with better outcomes have been observed in patients undergoing early CP,[8] the ideal timing for CP remains a subject of contention in the existing

ABBREVIATIONS: CP, cranioplasty; **DC,** craniectomy; **SSFS,** sinking skin flap syndrome; **TBI,** traumatic brain injury.

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literature, as highlighted by varying perspectives.[9,10] Notably, some studies have reported that the timing of CP is not correlated with certain complications, such as the risk of infection.[11,12]

Early procedures, within 3 months, were associated with improved neurological function outcomes.[13] Conversely, alternative studies support a delayed CP approach, suggesting intervention within 6 months after DC.[14,15] This delayed timing is justified by observations of good functional recovery and a reduced risk of complications. The absence of a standardized time frame underscores the ongoing debate and divergence in recommendations across various research findings.

To investigate the optimal timing for CP after severe TBI, we conducted a systematic review and meta-analysis of the current literature. We focused on complications and outcomes related to the timing and materials used in CP, aiming to provide a comprehensive understanding of the best practices for this procedure.

METHODS

This systematic review and meta-analysis was performed following the handbook of the Cochrane Collaboration and the Preferred Reporting Items for Systematic Reviews and Meta-Analysis statement guidelines.

Eligibility Criteria

Inclusion in this meta-analysis was restricted to studies that met all the following criteria: (1) randomized and observational studies, (2) reporting on intraprocedural or follow-up complications of CP after DC due to TBI, (3) comparing early and late CP procedures, and (4) studies reporting 4 or more patients. We excluded non-English articles, reviews, letters to the editor, abstracts, and commentaries from the initial assessment.

Search Strategy

We searched for complications after CP post-DC due to TBI across multiple databases, including PubMed, Embase, Cochrane, and Web of Science. The search strategy used terms related to "cranioplasty" or to "cranial defect," "calvarial defect" and "skull defect," such as "skull repair," "bone flap replacement" and "cranial replacement," coupled with keywords such as "traumatic brain injury," "TBI," "decompressive craniectomy," and "DC." A prospective registration was not performed for this study.

Screening

Search results were imported into Rayyan QCRI software for manual duplicate removal and initial screening based on titles and abstracts. This stage was conducted by 2 independent reviewers (L.B.P. and L.M.). Disagreements were resolved by the senior author (W.P.). The studies selected in this preliminary phase underwent full-text screening by 2 independent reviewers (L.M. and M.Y.F.). Conflicts were resolved by a third independent reviewer (L.B.P.).

Outcomes Definitions

The time between TBI and CP was divided as follows for the single-armed and two-armed analysis: "early" was considered for ≤35 days of surgical reconstruction and "late" for procedures happening after >35 days from TBI incident. Cranioplasties occurring after 3 months of TBI incident were considered as "ultra-late." Owing to the significant variation among studies in how they define a "complication," we established our definition as any postoperative situation necessitating further intervention. Major and minor complications were classified according to the following criteria: major complications were defined as being potentially life-threatening or if they brought permanent deficits to the patient, whereas minor complications were defined as not potentially life-threatening or not bringing permanent deficits to the patients. Data were sought regarding participant age and sex, publication date, number of patients receiving early CP, patients receiving late or ultra-late CP, overall complications, subdural fluid collection, infection rate, CP material, the severity of TBI, type of TBI/mechanism of injury, and mean operative time.

Quality Assessment

We assessed study quality using the Methodological Index for Non-Randomized Studies scale. This scale has 8 items for noncomparative studies and 4 for comparative studies. As our meta-analysis included only comparative studies, we used the full 12-item Methodological Index for Non-Randomized Studies scale. This allowed us to evaluate the methodological rigor and quality of the studies included in our analysis.

Statistical Analysis

A comparative analysis with odds ratio (OR) and 95% CIs was used, and P value <.05 was considered significant. A random effects model was adopted. I^2 statistics were used to assess for heterogeneity; $I^2 > 35\%$ was considered significant. Statistical analysis was performed using R Studio software (version 4.2.3, R Foundation for Statistical Computing). The Mantel-Haenszel method, the IV method, and the Restricted Maximum Likelihood were used.

RESULTS

Study Selection

A total of 3407 articles were identified through a search of multiple databases: 1028 in PubMed, 1685 in Embase, 653 in Web of Science, and 41 in Cochrane. Of these, 1112 non-duplicate citations were screened, excluding 981 articles because of their lack of relevance based on title and abstract review. The remaining 131 articles underwent full-text review, of which 118 were excluded because of the lack of description of procedure-related complications, the absence of TBI in the patients, or the inability to compare 2 different time intervals between DC and CP. Finally, 13 studies met the criteria for inclusion in this review and meta-analysis. One article was excluded after data extraction because it did not meet the required criteria. Thus, 12 studies were ultimately analyzed, and the details of the study selection process are illustrated in Figure 1.

Quality Assessment

The quality assessment of all 12 included studies is demonstrated below in Table 1.

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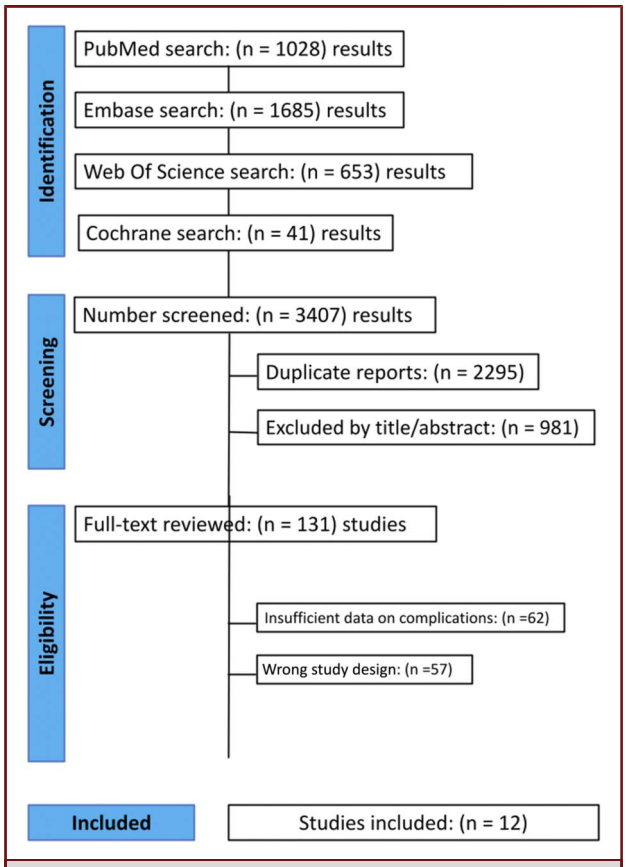

FIGURE 1. *Preferred Reporting Items for Systematic Reviews and Meta-Analysis flow diagram. This figure is original to this submission, so no credit or license is needed.*

Patients Baseline Characteristics

The analysis used data from 12 studies involving 1371 patients with TBI treated with CP. Each study had a patient range from 30 to 435. In adult-only cohort studies (n = 8, 66%), the mean patient age fell between the 4th and 6th decades of life. Operative times were reported in 33% of studies (n = 4), ranging from 59 to 180 minutes per surgery. TBI mechanisms were categorized based on various factors, including skull penetration, intracranial pressure, bleeding, and parenchymal complications, in 25% of studies (n = 3). TBI severity was assessed differently across studies, with 1 study classifying TBIs as "severe," 1 using the Extended Glasgow Outcome Scale, and the remaining 3 stratifying based on the Glasgow Coma Scale. All studies recorded cranioplasties as either "early" or "late," with Eaton (2022) including "Ultra early" and "intermediary" time cutoffs. The most common time cutoff for considering CP as "late" was 90 days (n = 5), with the earliest being 30 days (n = 1). Detailed patient baseline characteristics from the included studies are provided in Table 2.

Complications Assessment

Twelve studies were included in this meta-analysis, covering 1371 patients with TBI undergoing CP. Two studies involved both adult and pediatric populations, while the rest focused solely on adult patients (Table 2). Autologous bone tissue was the most commonly used CP material (66% of studies), followed by titanium mesh (25% of studies), and methyl methacrylate cement in 1 study (8% of studies). Among major complications, the most frequently reported was postoperative subdural fluid collection, affecting 132 patients (9% of patients), followed by hematoma in 81 (6% of patients), hydrocephalus in 65 (4% of patients), bone resorption in 38 (2% of patients), sinking skin flap syndrome (SSFS) in 31 (2% of patients), seizures in 28 (2% of patients), infection in 17 (1% of patients), wound healing complications in 9 (0.6% of patients), and fluid accumulation below the skin flap in 4 (0.2% of patients). Minor complications were described in 7 studies (58% of studies). The most frequently reported minor complication, although from 1 study only (Yang et al), was ventriculomegaly in 66 patients (38% of all minor complications), followed by 38 cases of bone resorption, 7 wound healing disturbances, 5 soft tissue injuries, 2 cerebrospinal fluid leaks, and 2 wound dehiscence. In an aggregate metric, considering multiple major and minor complications experienced by individual patients, 577 patient-level complications were reported (42% of patients), comprising 405 major (29.5% of patients) and 172 minor (12.5% of patients). Detailed major and minor complications per study are available in Table 3.

Pooled Analysis

Studies used different time cutoffs for defining "early" and "late" CP, with Eaton et al,[36], Tora et al,[3], and Zhang et al[35] using the longest cutoffs (over 90 days). Eaton et al[36] further divided this into "ultra-early" and "ultra-late" cohorts and outlined an institutional readiness assessment protocol. Comparing "early" and "late" cohorts, patients undergoing early CP had fewer subdural effusions (OR 0.3735 [0.1643; 0.8490], P = .0187, Figure 2), with no difference in infection rates (OR 0.7460 [0.2065; 2.6945], P = .6548, Figure 3).

Patients with < = 45 days post-TBI were analyzed separately, showing the following complication rates: fluid accumulation below skin flap: 0% ([0.00; 0.02]), hematoma: 3% ([0.00; 0.06], I^2 = 83%), hydrocephalus: 2% ([0.00; 0.04]), infection: 4% ([0.01; 0.07], I^2 = 68%), seizure: 2% ([0.00; 0.04]), and subdural effusion: 2% ([0.00; 0.04], I^2 = 84%). SSFS diagnosis ranged around 3% ([0.00; 0.06], I^2 = 83%). Aggregate complication rates were as follows: total complications: 26% ([0.20; 0.32], I^2 = 96%), major complications: 24% ([0.20; 0.28], I^2 = 99%), and minor complications: 1% ([0.00; 0.03], I^2 = 78%).

The same analysis was conducted for all CP patients until the "ultra-late" cutoff (up to 3 months post-TBI), revealing the following proportions: fluid accumulation below skin flap: 0% ([0.00; 0.01]), hematoma presence: 2% ([0.01; 0.03], I^2 = 66%), hydrocephalus presence: 2% ([0.00; 0.03]), infection presence:

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TABLE 1. Quality Assessment of the Included Studies Using the Methodological Index for Non-Randomized Studies Scale

Minors criteria	Chun 2011[30]	Eaton 2022[36]	Jiang 2020[31]	Kumar 2018[33]	Morgan 2023[29]	Piedra 2014[10]	Schuss 2012[18]	Song 2013[19]	Tora 2021[3]	Vyas 2021[34]	Yang 2018[32]	Zhang 2010[35]
1. A clearly stated aim: the question addressed should be precise and relevant in the light of available literature	2	2	2	2	2	2	2	2	2	2	2	2
2. Inclusion of consecutive patients: all patients potentially fit for inclusion (satisfying the criteria for inclusion) have been included in the study during the study period (no exclusion or details about the reasons for exclusion)	2	2	2	2	0	2	1	2	2	2	2	2
3. Prospective collection of data: data were collected according to a protocol established before the beginning of the study	0	0	2	2	0	0	2	0	0	0	0	0
4. End points appropriate to the aim of the study: unambiguous explanation of the criteria used to evaluate the main outcome, which should be in accordance with the question addressed by the study. In addition, the end points should be assessed on an intention-to-treat basis	2	2	2	1	2	2	2	1	2	2	2	2
5. Unbiased assessment of the study end point: blind evaluation of objective end points and double-blind evaluation of subjective end points. Otherwise, the reasons for not blinding should be stated	0	0	0	0	0	0	0	0	0	0	0	0
6. Follow-up period appropriate to the aim of the study: the follow-up should be sufficiently long to allow the assessment of the main end point and possible adverse events	2	0	0	2	2	1	0	2	2	1	2	2
7. Loss to follow-up less than 5%: all patients should be included in the follow-up. Otherwise, the proportion lost to follow-up should not exceed the proportion experiencing the major end point	1	0	2	2	2	2	0	2	2	1	2	2
8. Prospective calculation of the study size: information of the size of detectable difference of interest with a calculation of 95% confidence interval, according to the expected incidence of the outcome event, and information about the level for statistical significance and estimates of power when comparing the outcomes	1	2	2	2	1	1	2	1	2	1	1	1

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TABLE 1. Continued.

Minors criteria	Chun 2011[30]	Eaton 2022[36]	Jiang 2020[31]	Kumar 2018[33]	Morgan 2023[29]	Piedra 2014[10]	Schuss 2012[18]	Song 2013[19]	Tora 2021[3]	Vyas 2021[34]	Yang 2018[32]	Zhang 2010[35]
9. An adequate control group: having a gold standard diagnostic test or therapeutic intervention recognized as the optimal intervention according to the available published data	2	1	2	1	1	1	1	1	1	1	1	2
10. Contemporary groups: control and studied group should be managed during the same time period (no historical comparison)	1	1	1	2	1	1	1	2	1	1	1	1
11. Baseline equivalence of groups: the groups should be similar regarding the criteria other than the studied end points. Absence of confounding factors that could bias the interpretation of the results	2	2	2	2	2	2	2	2	1	2	2	2
12. Adequate statistical analyses: whether the statistics were in accordance with the type of study with calculation of confidence intervals or relative risk	0	2	2	0	0	1	2	0	2	0	0	0
Sum of the criteria	15	14	19	18	13	15	15	15	17	13	15	16

4% ([0.02; 0.05], I^2 = 59%), seizure occurrence: 1% ([0.00; 0.02]), and subdural effusion presence: 0% ([0.00; 0.01], I^2 = 65%). SSFS diagnosis ranged around 0% ([0.00; 0.01]). Aggregate complication rates were 19% for total complications ([0.16; 0.22], I^2 = 93%), 17% for major complications ([0.15; 0.20], I^2 = 98%), and 0% for minor complications ([0.00; 0.01], I^2 = 73%).

The patient cohort was divided into subgroups based on the type of graft used: autologous and nonautologous. For autologous grafts, fluid accumulation below the skin flap was 0% ([0.00; 0.02]), the presence of hematoma was 1% ([0.00; 0.03]), and the presence of hydrocephalus was 0% ([0.02; 0.02]). The presence of infection was 3% ([0.00; 0.06], I^2 = 59%; Figure 4). The occurrence of seizure was 0% ([0.00; 0.01]), and the presence of subdural effusion was 0% ([0.00; 0.02], I^2 = 0%; Figure 5). The development of SSFS was 0% ([0.00; 0.01]). Total complications were 17% ([0.12; 0.21], I^2 = 93%; Figure 6), with minor complications at 2% ([0.00; 0.05], I^2 = 88%; Figure 7) and major complications at 11% ([0.07; 0.16], I^2 = 74%; Figure 8). Bone resorption was observed at 2% ([0.00; 0.04], I^2 = 75%).

For nonautologous grafts, fluid accumulation below the skin flap was 1% ([0.00; 0.04]). The presence of hematoma was 0% ([0.00; 0.03]—no cases), and the presence of hydrocephalus was 0% ([0.00; 0.03]—no cases). The presence of infection was 0% ([0.00; 0.03]—no cases; Figure 9). The occurrence of seizure was 1% ([0.00; 0.04]), and the presence of subdural effusion was 3% ([0.00; 0.07], I^2 = 22%; Figure 10). The development of SSFS was 0% ([0.00; 0.03]—no cases). Total complications were 8% ([0.02; 0.14], I^2 = 0%; Figure 11), with minor complications at 3% ([0.00; 0.07], I^2 = 29%; Figure 12) and major complications at 2% ([0.00; 0.05], I^2 = 13%; Figure 13). More detailed information on for parameters obtained for each subgroup on Table 4:

Finally, all CP cohorts were compared with the 3-month "ultra-late" cutoff. Cranioplasties performed within 3 months of the TBI incident were associated with an increased risk of total complications (OR 1.4, P = .02, Figure 14). Paradoxically, they were linked to fewer minor complications (OR 0.4, P = .008, Figure 15). No differences were observed in major complications (Figure 16), infection rate, subdural effusion, presence of seizures, or hematoma, compared with the ultra-late cohort. In addition, cranioplasties within 3 months were associated with an increased risk of hydrocephalus (OR 3.2, P = .003, Figure 17).

DISCUSSION

This review incorporates 12 studies, involving 1371 adult and pediatric patients with TBI undergoing CP. However, limited high-quality evidence addresses key clinical questions. CP, using either a preserved autologous bone flap or synthetic prosthesis, can restore a degree of mechanical protection to the brain. It aims to

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TABLE 2. Baseline Characteristics of Included Studies

Study	Study type	Adult or pediatric (A or P)	Number of patients with TBI	Mean age	Time of cranioplasty cutoffs	Number of patients per time frame of cranioplasty: ultra-early/early/intermediate/late	Mean operative time	Type of TBI/ mechanism of injury	Severity of TBI	Cranioplasty material
Chun 2011[30]	R	A/P	45	≤1 mo: 49.33 years. Range: 7-77 >3 mo: 49 years. Range: 28-68	Early: ≤1 mo Late: >3 mo	0/30/0/15	≤1 mo: 95.33 min (range: 85-110) >3 mo: 133.67 min (range: 110-180)	≤1 mo: 18 ASDH, 9 TIPH, 3 AEDH + massive cerebral contusion >3 mo: 6 ASDH, 9 TIPH	NR	MMC cement
Eaton 2022[36]	R	A	435	<42 days: 54.2 ± 21.3 years ≤90 days: 51.1 ± 18.3 years 91-180 days: 50.7 ± 15.2 years >180 days: 52.5 ± 15.4 years	Ultra-early: <42 days Early: ≤90 days Intermediate: 91-180 days Late: >180 days	56/141/187/107	NR	NR	NR	Autologous or synthetic polyether ether ketone implant
Jiang 2020[31]	P	A	60	Range: 17-65 years	Early: 4-6 weeks Late: 3-6 months	0/28/0/32	NR	NR	Severe TBI	NR
Kumar 2018[33]	P	A/P	42	41.40 ± 15.95 years Range: 16-70	Early: ≤2 mo Late >2 mo	0/21/0/21	NR	NR	GOSE scale preoperative ≤2 mo: 4 severe, 16 moderate, 1 lower good (median GOSE 5) >2mo: 21 severe (median GOSE 4)	Autologous

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TABLE 2. Continued.

Study	Study type	Adult or pediatric (A or P)	Number of patients with TBI	Mean age	Time of cranioplasty cutoffs	Number of patients per time frame of cranioplasty: ultra-early/early/intermediate/late	Mean operative time	Type of TBI/mechanism of injury	Severity of TBI	Cranioplasty material
Morgan 2023[29]	R	P	30	≤30 days: 6.84 ± 6.30 years >30 days: 9.61 ± 7.58 years	Early: ≤30 days Late: >30 days	0/24/0/6	NR	≤30 days—Original insult: 10 closed head injury, 9 skull fracture, 5 penetrating head wound Mechanism of injury: 6 motor vehicle collision, 6 known accidental, 7 nonaccidental trauma, 5 GSW >30 days—Original insult: 2 closed head injury, 3 skull fracture, 1 penetrating head wound Mechanism of injury: 4 motor vehicle collision, 1 nonaccidental trauma, 1 GSW	NR	Autologous
Piedra 2014[10]	R	A/P	157	<12 weeks: 27.2 ± 20 years ≥12 weeks: 31.9 ± 17.1 years	Early: <12 weeks Late: ≥12 weeks	0/78/0/79	<12 weeks: 102 ± 45 min ≥12 weeks: 125 ± 71 min	<12 weeks: 41 SDH, 24 elevated ICP, 7 EDH, 5 SDH/EDH, 1 skull fracture ≥12 weeks: 37 SDH, 32 elevated ICP, 8 EDH, 2 SDH/EDH	NR	<12 weeks: 78 autologous ≥12 weeks: 68 autologous, 11 synthetic
Schuss 2012[18]	R	A	98	38 ± 19 years	Early: ≤2 mo Late: >2 mo	0/29/0/69	124 ± 36 min	NR	NR	Autologous

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TABLE 2. Continued.

Study	Study type	Adult or pediatric (A or P)	Number of patients with TBI	Mean age	Time of cranioplasty cutoffs	Number of patients per time frame of cranioplasty: ultra-early/early/intermediate/late	Mean operative time	Type of TBI/mechanism of injury	Severity of TBI	Cranioplasty material
Song 2013[19]	R	A	43	<12 weeks: 40.1 ± 16.52 years ≥12 weeks: 44.1 ± 15.30 years	Early: <12 weeks Late: ≥12 weeks	0/25/0/18	NR	NR	Preoperative GCS <12 weeks: ≤8: 8, 8-12: 8, >12: 9 (mean 10.0 ± 3.01) ≥12 weeks: ≤8: 7, 8-12: 3, >12: 8 (mean 9.8 ± 3.78)	Titanium mesh
Tora 2021[3]	R	A	142	NR	Early: ≤90 days Late: >90 days	0/51/0/91	NR	NR	NR	Autologous, prosthesis
Vyas 2021[34]	R	A	90	≤2 mo: 41.75 ± 13.4 years >2 mo: 45.61 ± 12.5 years	Early: ≤2 mo Late: >2 mo	0/44/0/46	≤2 mo: 59.39 ± 7.3 min >2 mo: 77.28 ± 9.9 min	NR	NR	≤2 mo: 44 autologous >2 mo: 44 autologous, 2 artificial
Yang 2018[32]	R	A	159	53.01 ± 16.44 years	Early: <35 days Late: ≥35 days	0/15/0/144	NR	NR	Initial GCS (mean 8.33 ± 3.46) ≤8: 84 8-12: 45 >12: 30	Autologous or allograft bone tissue
Zhang 2010[35]	R	A	70	≤3 mo: 38.7 ± 12.6 years >3 mo: 41.4 ± 13.3 years (NDS), 39.1 ± 13.7 years (DS)	Early: ≤3 mo Late: >3 mo	0/23/0/47	NR	NR	Severe TBI (GCS 5-8 preop) Mean preoperative GCS score ≤3 mo: 6.6 ± 1.1 >3 mo: 6.3 ± 1.0 DS, 6.4 ± 1.1 NDS	Titanium mesh

AEDH, acute epidural hematoma; ASDH, acute subdural hematoma; DS, dura suture; EDH, extradural hemorrhage; GSW, gunshot wound; ICP, intracranial pressure; MMC, methyl methacrylate; NDS, nondura suture; SDH, subdural hemorrhage; TIPH, traumatic intraparenchymal hemorrhage.

Early: between 15 and 30 days; P: prospective ultra-early: <15 days; R, retrospective; Intermediate: between 30 and 90 days; Late: > 90 days.

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TABLE 3. Major and Minor Complications Reported in Selected Studies

Study	Overall complication rate	Total N of complications ultra-early group	Total N of complications early group	Total N of complications intermediate group	Total N of complications late group	Major/minor complications in the "early" cohort	Major/minor complications in the "late" cohort	Complications major/minor by the time of cranioplasty
Chun 2011[30]	22/45	NA	2/30	NA	20/15	0/2	8/12	≤1 mo—major: NR Minor: 2 subdural fluid collections >3 mo—major: 1 infection, 7 subdural fluid collection Minor: 5 soft tissue injury, 2 inadequate dissection
Eaton 2022[36]	73/435	11/56	25/141	22/187	20/107	Ultra-early: 11/0 Early: 26/0	Intermediate: 25/0 Late: 22/0	<42 days—major: 3 infections, 3 seizures, 3 hematoma, 2 hydrocephalus Minor: NR ≤90 days—major: 5 hydrocephalus, 5 seizure, 5 hematoma, 11 infection Minor: NR 91-180 days—major: 5 hydrocephalus, 7 seizure, 6 hematoma, 7 infection Minor: NR >180 days—major: 2 hydrocephalus, 8 seizure, 4 hematoma, 8 infection Minor: NR
Jiang 2020[31]	46/60	NA	21/28	NA	25/32	21/0	25/0	4-6 weeks—major: 6 infection, 5 fluid accumulation, 5 scalp necrosis, 5 intracranial hematoma Minor: NR 3-6 months—major: 6 intracranial infection, 5 fluid accumulation, 7 scalp necrosis, 7 intracranial hematoma Minor: NR
Kumar 2018[33]	3/42	NA	3/21	NA	0/21	3/0	0/0	≤2 mo—major: 1 EDH, 1 seizure, 1 ICH Minor: NR >2 mo—major: NR Minor: NR

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TABLE 3. Continued.

Study	Overall complication rate	Total N of complications ultra-early group	Total N of complications early group	Total N of complications intermediate group	Total N of complications late group	Major/minor complications in the "early" cohort	Major/minor complications in the "late" cohort	Complications major/minor by the time of cranioplasty
Morgan 2023[29]	19/30	NA	16/24	NA	3/6	7/9	0/3	≤30 days—major: 5 infection, 1 hydrocephalus, 1 subdural fluid collection Minor: 3 wound healing disturbance, 3 bone resorption, 2 CSF leakage, 1 subgaleal fluid collection >30 days—major: NR Minor: 1 wound healing disturbance, 2 bone resorption
Piedra 2014[10]	55/157	NA	27/78	NA	28/79	15/12	13/15	<12 weeks—major: 3 hematoma, 6 infection, 6 hydrocephalus Minor: 12 bone graft resorption ≥12 weeks—major: 1 hematoma, 11 infection, 1 hydrocephalus Minor: 15 bone graft resorption
Schuss 2012[18]	21/98	NA	7/29	NA	14/69	NR	NR	NR
Song 2013[19]	5/43	NA	2/25	NA	3/18	2/0	3/0	<12 weeks—major: 2 subdural fluid collection Minor: NR ≥12 weeks—major: 3 subdural fluid collection Minor: NR
Tora 2021[3]	47/142	NA	31/51	NA	16/91	29/2	11/5	≤90 days—major: 5 infection, 10 reoperation, 8 hydrocephalus, 6 extra-axial collection Minor: 1 dehiscence, 1 bone resorption >90 days—major: 2 infection, 6 reoperation, 2 hydrocephalus, 1 extra-axial collection Minor: 1 dehiscence, 4 bone resorption

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TABLE 3. Continued.

Study	Overall complication rate	Total N of complications ultra-early group	Total N of complications early group	Total N of complications intermediate group	Total N of complications late group	Major/minor complications in the "early" cohort	Major/minor complications in the "late" cohort	Complications major/minor by the time of cranioplasty
Vyas 2021[34]	9/90	NA	2/44	NA	7/46	2/0	6/1	≤2 mo—major: 2 infection Minor: NR >2 mo—major: 1 hematoma, 3 hydrocephalus, 2 sunken brain Minor: 1 bone resorption
Yang 2018[32]	275/159	NA	17/15	NA	258/144	15/2	194/64	<35 days—major: 2 sinking flap syndrome, 1 hydrocephalus, 8 ipsilateral or interhemispheric hygroma, 3 contralateral hematoma or hygroma, 1 revision because of infection Minor: 2 ventriculomegaly ≥35 days—major: 27 sinking skin flap syndrome, 29 hydrocephalus, 92 ipsilateral or interhemispheric hygroma, 29 contralateral hematoma or hygroma, 17 revision because of infection Minor: 64 ventriculomegaly
Zhang 2010[35]	12/70	NA	3/23	NA	9/47	1/2	5/4	≤3 mo—major: 1 seizure Minor: 1 wound healing disturbance, 1 fluidity below skin flap >3 mo—major: 2 perioperative meninges breakage, 3 seizure Minor: 2 fluidity below skin flap, 2 wound healing disturbance

"Overall complication rate" is defined as the "Total number of complications/Total number of patients".

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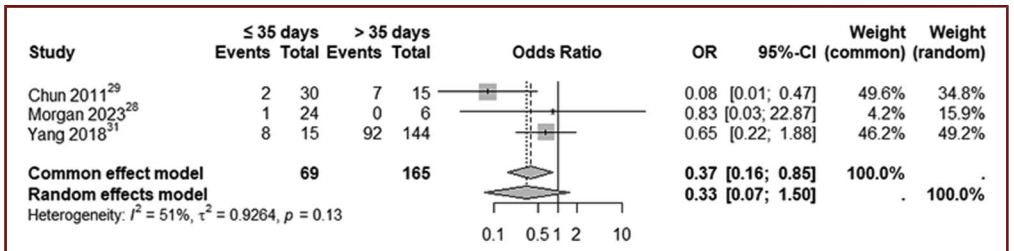

FIGURE 2. *Early vs late cranioplasty (35 days as cutoff) comparative analysis for subdural effusion. This figure is original to this submission, so no credit or license is needed.*

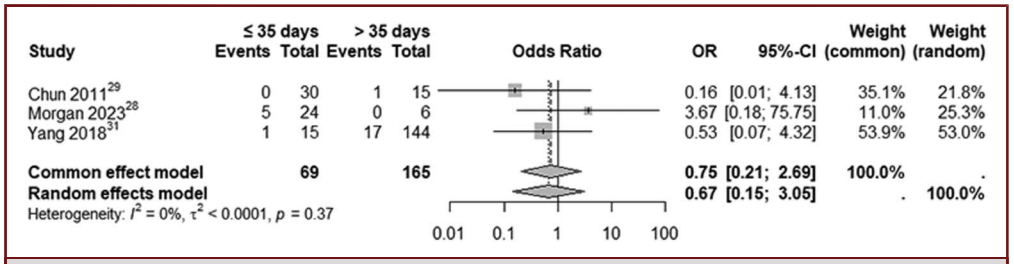

FIGURE 3. *Early vs late cranioplasty (35 days as cutoff) comparative analysis for infection. This figure is original to this submission, so no credit or license is needed.*

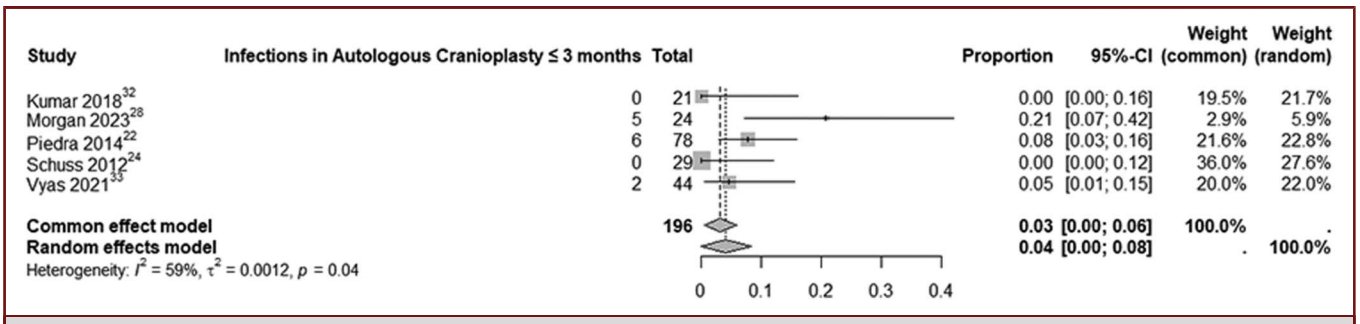

FIGURE 4. *<90 days single-arm meta-analysis of infection in patients receiving autologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

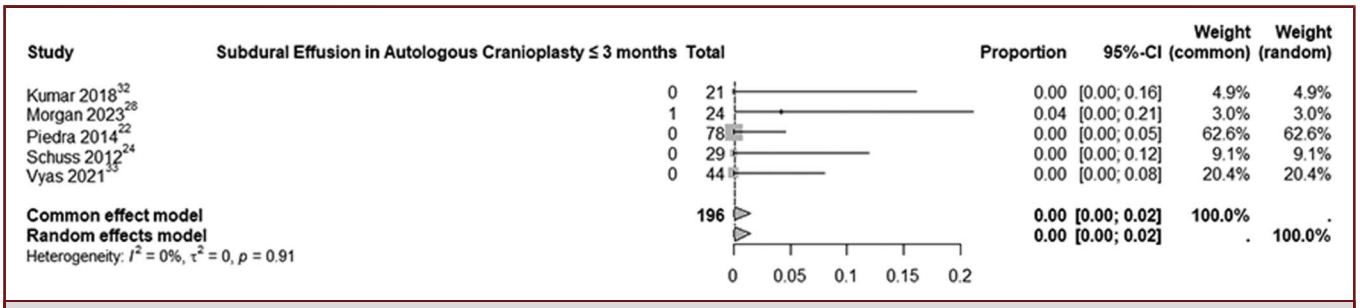

FIGURE 5. *<90 days single-arm meta-analysis of subdural effusion in patients receiving autologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

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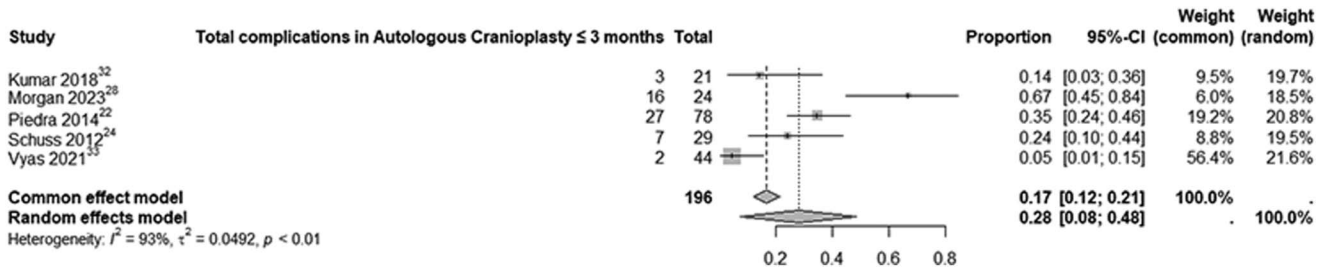

FIGURE 6. *<90 days single-arm meta-analysis of total number of complications in patients receiving autologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

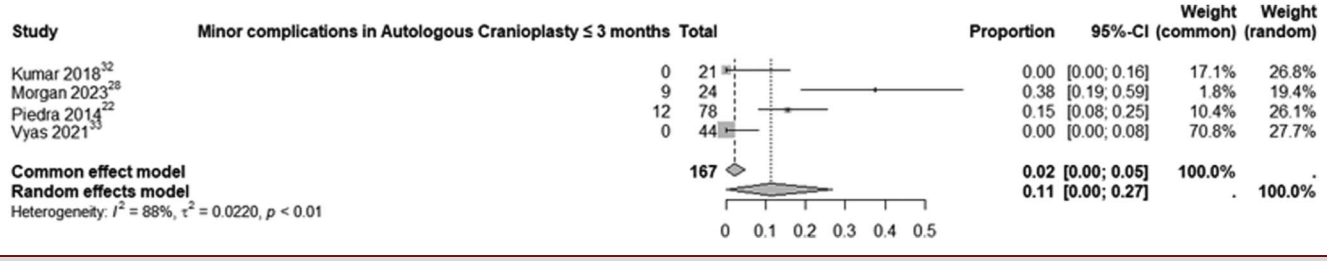

FIGURE 7. *<90 days single-arm meta-analysis of minor complications in patients receiving autologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

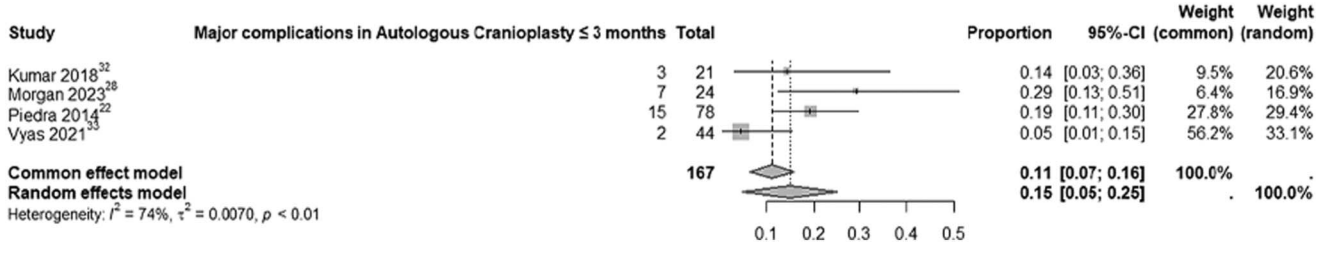

FIGURE 8. *<90 days single-arm meta-analysis of major complications in patients receiving autologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

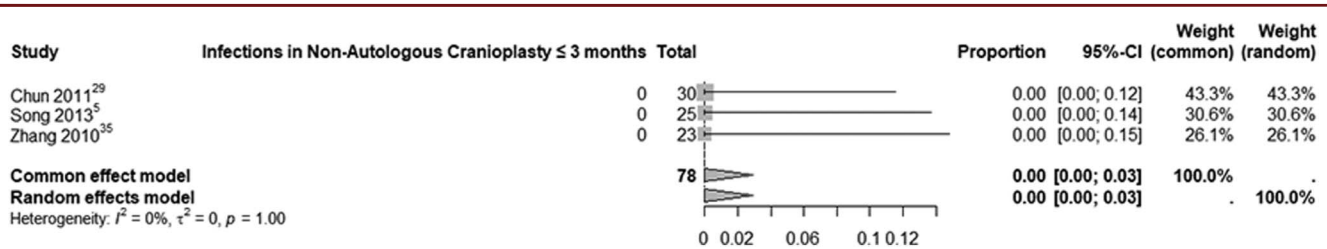

FIGURE 9. *<90 days single-arm meta-analysis of infection in patients receiving nonautologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

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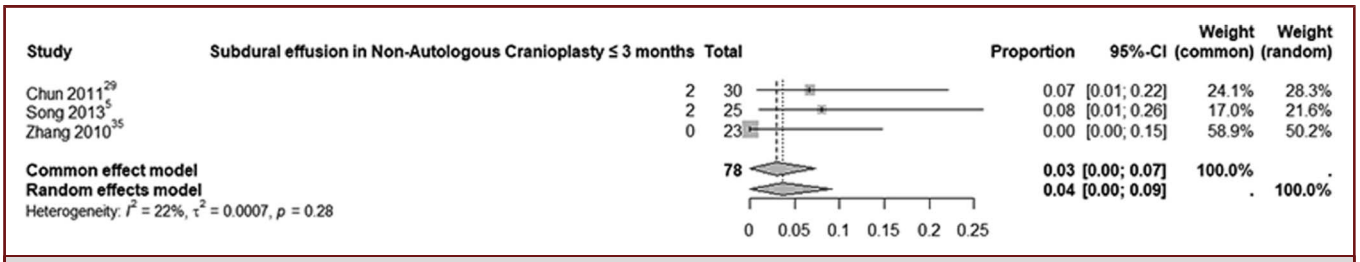

FIGURE 10. *<90 days single-arm meta-analysis of subdural effusion in patients receiving nonautologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

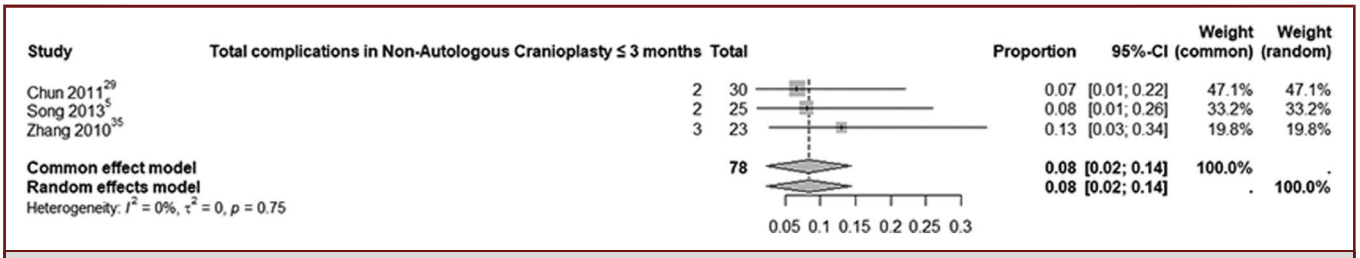

FIGURE 11. *<90 days single-arm meta-analysis of total number of complications in patients receiving nonautologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

diminish the risk of falls by enhancing vestibular system equilibrium, addressing the syndrome of trephined, and simultaneously improving craniofacial aesthetics and neurological performance, particularly regarding the magnitude of change in early indications postcraniectomy.[8,16-18]

Complications vs Timing

In general, reported complication rates from CP, regardless of timing, range from 15% to 30%,[22] and a comprehensive systematic review revealed a pooled overall rate of 19.5% for all CP indications and a rate of 25.6% for TBI indications.[21] In our

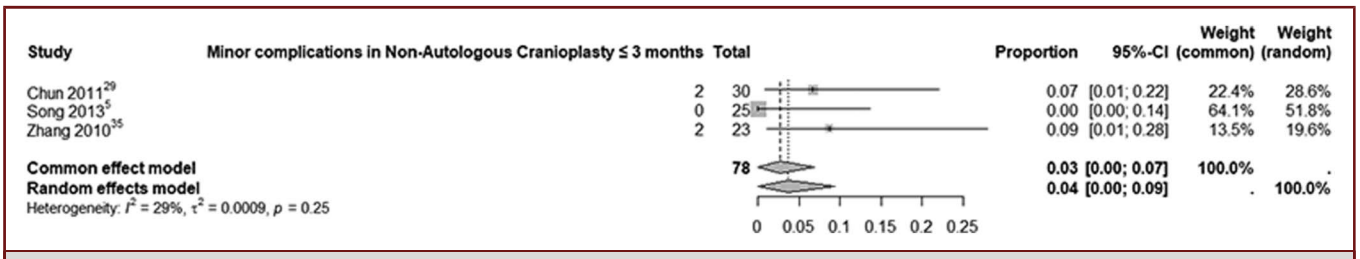

FIGURE 12. *<90 days single-arm meta-analysis of minor complications in patients receiving nonautologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

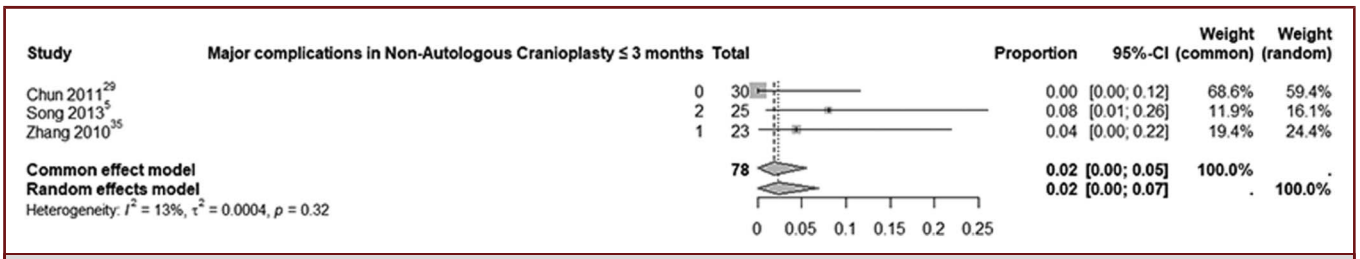

FIGURE 13. *<90 days single-arm meta-analysis of major complications in patients receiving nonautologous cranioplasty. This figure is original to this submission, so no credit or license is needed.*

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TABLE 4. Autologous × Nonautologous Findings

Autologous grafts	Nonautologous grafts
Fluid accumulation below skin flap: 0% ([0.00; 0.02])	Fluid accumulation below skin flap: 1% ([0.00; 0.04])
Presence of hematoma: 1% ([0.00; 0.03])	Presence of hematoma: 0% ([0.00; 0.03]—no cases)
Presence of hydrocephalus: 0% ([0.02; 0.02])	Presence of hydrocephalus: 0% ([0.00; 0.03]—no cases)
Presence of infection: 3% ([0.00; 0.06], I^2 = 59%; Figure 4)	Presence of infection: 0% ([0.00; 0.03]—no cases; Figure 9)
Occurrence of seizure: 0% ([0.00, 0.01])	Occurrence of seizure: 1% ([0.00, 0.04])
Presence of subdural effusion: 0% ([0.00; 0.01], I^2 = 0%; Figure 5)	Presence of subdural effusion: 3% ([0.00; 0.07], I^2 = 22%; Figure 10)
Development of Sinking Skin Flap Syndrome: 0% ([0.00; 0.01])	Development of Sinking Skin Flap Syndrome: 0% ([0.00; 0.03]—no cases)
Total complications: 17% ([0.12; 0.21], I^2 = 93%; Figure 6)	Total complications: 8% (0.02; 0.14], I^2 = 0%; Figure 11)
Minor complications: 2% ([0.00; 0.05], I^2 = 88%; Figure 7)	Minor complications: 3% ([0.00; 0.07], I^2 = 29%; Figure 12)
Major complications: 11% ([0; 07; 0.16], I^2 = 74%; Figure 8)	Major complications: 2% ([0; 00; 0.05], I^2 = 13%; Figure 13)
Bone resorption: 2% ([0.00; 0.04], I^2 = 75%)	—

analysis, the total complication rate was 26.9%, and considering multiple complications experienced by individual patients, 1179 of 1371 patient-level complications were reported, representing 86% of the patients. Frequently encountered complications encompass infection, intracranial hemorrhage, extra-axial fluid collections, hydrocephalus, seizures, and bone resorption.

The optimal timing for CP remains highly variable.[25] Traditionally, it has been suggested that performing CP around 6 months postinjury minimizes the risk of infection[26,27] or complications[20] after DC. By contrast, Oh et al[28] observed significantly higher infection rates in the late CP group. Recent research challenges the idea that a 6-month delay is necessary to minimize the infection risk, and these studies show no statistical difference in infection rates and most complications between different timing intervals besides an increase in odds of hydrocephalus with early CP.[21-23] The timing varied widely between the TBI indication series, with different cutoffs ranging from 30 days,[29-31] 35 days,[32] 2 months,[9,33,34] and 3 months[3,19,35] as relative for ultra-early and early descriptions. Only Eaton et al[36] followed the proposed time frame (post-craniotomy) cutoffs by the consensus meeting on post-traumatic CP[25]: ultra-early CP is up to 6 weeks after craniectomy, early CP is 6 weeks to 3 months, intermediate 3 to 6 months, and delayed more than 6 months. Although they recognize that clinical status should be considered when deciding the timing of CP, it can be helpful for clinical benchmarking and research purposes.[25]

Our ultra-early analysis revealed a significant decrease in the odds of subdural effusion with an earlier approach, while no difference in the odds of infection was observed. These findings

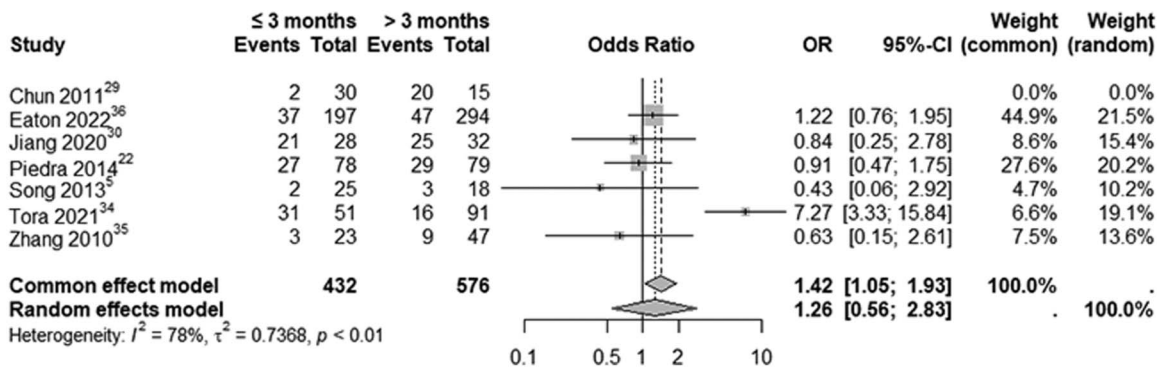

FIGURE 14. *<90 days vs >90 days cutoff of cranioplasty for total complications. This figure is original to this submission, so no credit or license is needed.*

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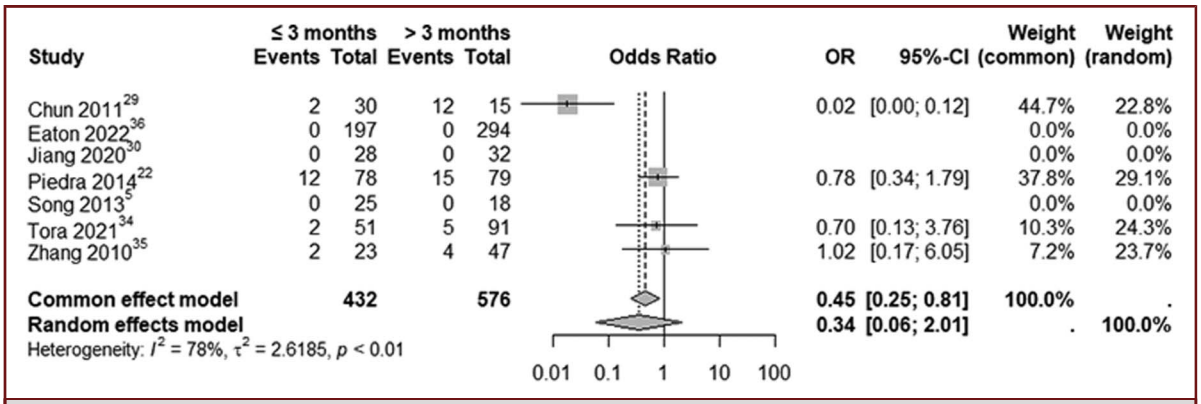

FIGURE 15. *<90 days vs >90 days cutoff of cranioplasty for minor complications. This figure is original to this submission, so no credit or license is needed.*

are consistent with previous results, although previous studies combined reports with longer time frame cutoffs,[21,24] suggesting that even earlier approaches have little impact on infection rates but may offer benefits in reducing fluid collection.

We defined cutoff timing as 35 days, relying on the cutoff established by Yang et al,[32] which carried significant weight in our analysis. Patient-level data were available from Chun and colleagues,[30] revealing that none in the control group underwent CP in fewer than 35 days. In their cohort, ultra-early surgery facilitated intraoperative dissection plane definition without additional complications, such as infection, subdural hygroma, or brain parenchymal hemorrhage, as observed in the control group. The only limitation arises from Morgan et al,[29] who did not specify the exact date for the control group undergoing late CP. However, the mean timing for their 6 control patients was 86.83 days (SD = 44.82), minimizing the risk of erroneously categorizing any of these patients within the 30 to 35-day interval.

In our assessment comparing early with intermediate and delayed CP (within 3 months vs more than 3 months), we observed a higher association with hydrocephalus in the earlier approach, with no differences in infection rate, seizures, or hematoma, consistent with previous studies.[21,24,37] However, with an expanded study pool, no difference was noted in the odds of subdural effusion, contradicting previous systematic reviews.[21] Furthermore, the earlier approach was linked to fewer minor complications, with no discernible differences in major complications, but posed an elevated risk of total complications. Similarly, Schuss et al[9] related that the early group suffered significantly more complications (25.9%) compared with the delayed group (14.2%) in their study, besides there was no description of specific complications.

Autograft vs Allograft

Autologous bone is frequently favored over allograft devices due to its biological characteristics, optimal compatibility with the

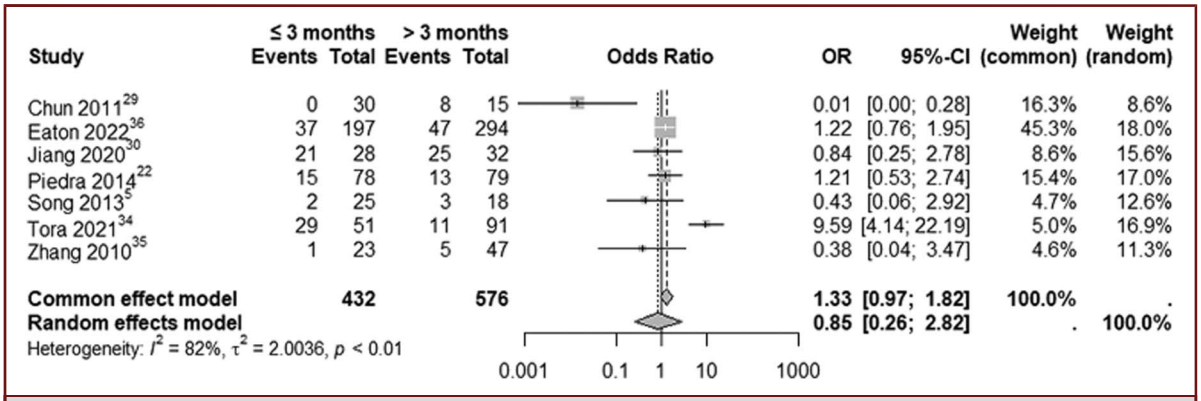

FIGURE 16. *<90 days vs >90 days cutoff of cranioplasty for major complications. This figure is original to this submission, so no credit or license is needed.*

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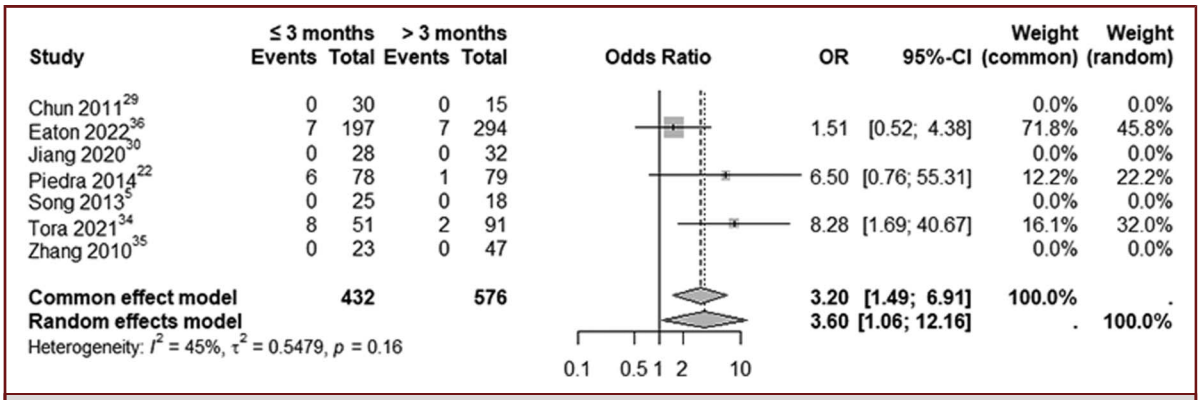

FIGURE 17. *<90 days vs >90 days cutoff of cranioplasty for hydrocephalus. This figure is original to this submission, so no credit or license is needed.*

bony defect, and lower cost.[38] By contrast, allograft has traditionally been considered a secondary option because of perceived higher risks of infection and rejection and associated costs,[39] despite recent materials showing potential for superior cosmetic results.[40,41] Notably, a previous meta-analysis found no significant difference in infection rates.[32]

The literature links few adverse events directly to these materials, besides bone flap resorption which is a complication specifically associated with autologous CP.[42-46] In adults, the reported incidence of bone flap resorption ranges from 2% to 32%,[42,43] and in pediatric patients, the risk is higher, with rates increasing to 50% to 66%.[44,45] Our systematic review, however, reported a lower pooled incidence of bone flap resorption at 4%. Notably, only 2 studies with pediatric populations reported this outcome: 17.2% in Piedra et al[10] and 16.7% in Morgan et al,[29] both irrespective of timing.

Limitations

Our study has limitations worth noting. First, the lack of randomized studies is a constraint, as most studies are retrospective and single-center. In addition, significant heterogeneity among the included studies introduces biases because of differences in methodologies, patient populations, and timing definitions, potentially leading to varied outcomes. Moreover, there is a shortage of studies involving pediatric populations, with limited subgroup analysis when present in mixed studies, emphasizing the need for more research in this demographic for comprehensive conclusions. Studies' differences in timing categorization of cranioplasties in different cohorts limited the number of possible definitions for our analysis; however, the 30-35 days window seemed to divide the timing for analysis functionally because this interval was not tangential but a single study. In addition, the functional benefits of CP timing were not specifically assessed, representing a gap in our exploration. Last, the classification between major and minor complications was not made entirely clear by the included studies. So, there is an interpretative component that in some cases may over or underestimate the results.

CONCLUSION

Despite the established significance of CP, the review reveals a lack of high-quality evidence addressing key clinical questions, such as optimal timing, choice of material, and long-term outcomes. Complications after CP, regardless of timing, remain a significant concern, with infection, intracranial hemorrhage, and bone resorption among the frequently encountered issues, and our analysis evidenced that a single patient may frequently experience multiple of those complications during the treatment course. The optimal timing for CP remains variable mainly due to the heterogeneity in the studies. This variability challenges the traditional notion of a 6-month delay to minimize the infection risk. Some studies suggest that earlier CP cohorts may have a lower risk of fluid recollection, while infection rates are comparable between early and late CP cohorts. Although current literature has followed a trend to favor earlier cranioplasties, we have not found evidence to support the claim. The intrinsic bias in the observational studies, especially regarding the definition of early and late cranioplasties, may have played a significant role in this matter. Furthermore, the comparison between autograft and allograft materials highlights the complexities involved in material selection, with autologous bone tissue favored for its biological characteristics despite the risk of bone flap resorption.

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The authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

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